

Topic: Public Sentiment on World War 3

Discussions on Twitter

This assignment focuses on performing sentiment analysis on tweets related to World War 3 discussions. The topic was selected because it is widely discussed on social media platforms where people express fear, opinions, and reactions regarding global conflicts.

Sentiment analysis helps in understanding public opinion by classifying text into positive, negative, and neutral categories.

DATA COLLECTION & PREPROCESSING :

A dataset of 100 tweets related to World War 3 was collected manually from Twitter using search keywords such as “World War 3”, “WW3”, and “global conflict”.

Each tweet was manually labeled into one of three sentiment categories: positive, negative, or neutral based on its content and tone.

The following preprocessing steps were applied:

- Conversion of text to lowercase
- Removal of URLs and mentions
- Removal of punctuation and special characters
- Removal of extra spaces

These steps helped in cleaning the text data and improving the performance of the machine learning models.

DATA SPLITTING :

The dataset was divided into training and testing sets using an 80:20 ratio.

- 80 tweets were used for training the models
- 20 tweets were used for testing the models

This split ensures that the model is trained on sufficient data while also being evaluated on unseen data.

MODEL TRAINING & CLASSIFICATION :

Three machine learning classifiers were used for sentiment analysis:

- Naive Bayes
- Support Vector Machine (SVM)
- Logistic Regression

The cleaned text data was converted into numerical form using TF-IDF (Term Frequency-Inverse Document Frequency) vectorization.

Each model was trained on the training dataset and used to predict sentiments on the testing dataset.

MODEL EVALUATION

The performance of each model was evaluated using precision and recall metrics.

The results are as follows:

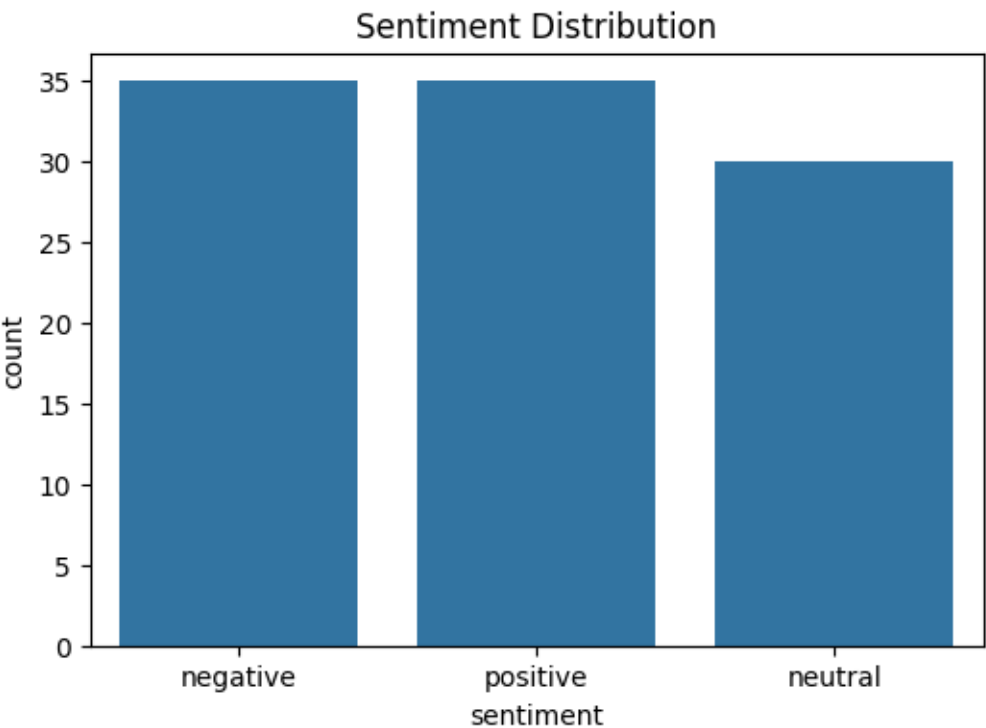
	Model	Precision	Recall	Accuracy
0	Naive Bayes	1.0	1.0	1.0
1	SVM	1.0	1.0	1.0
2	Logistic Regression	1.0	1.0	1.0

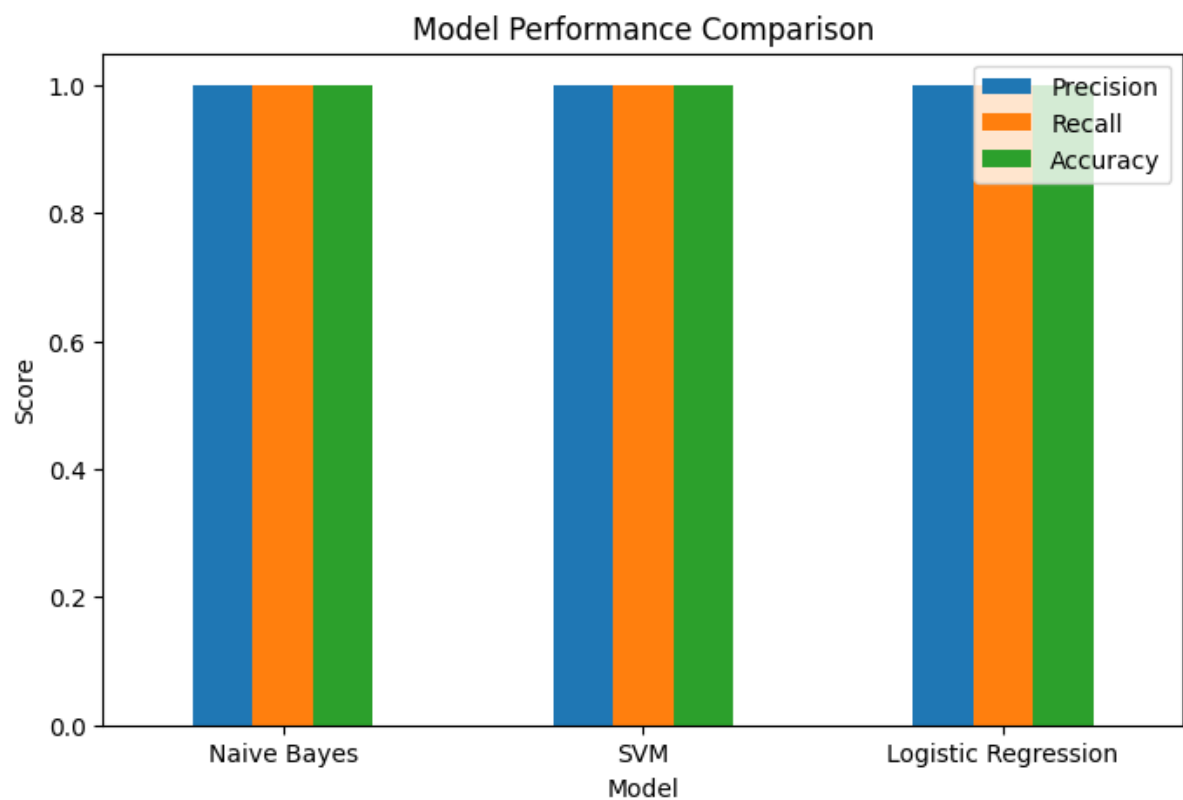
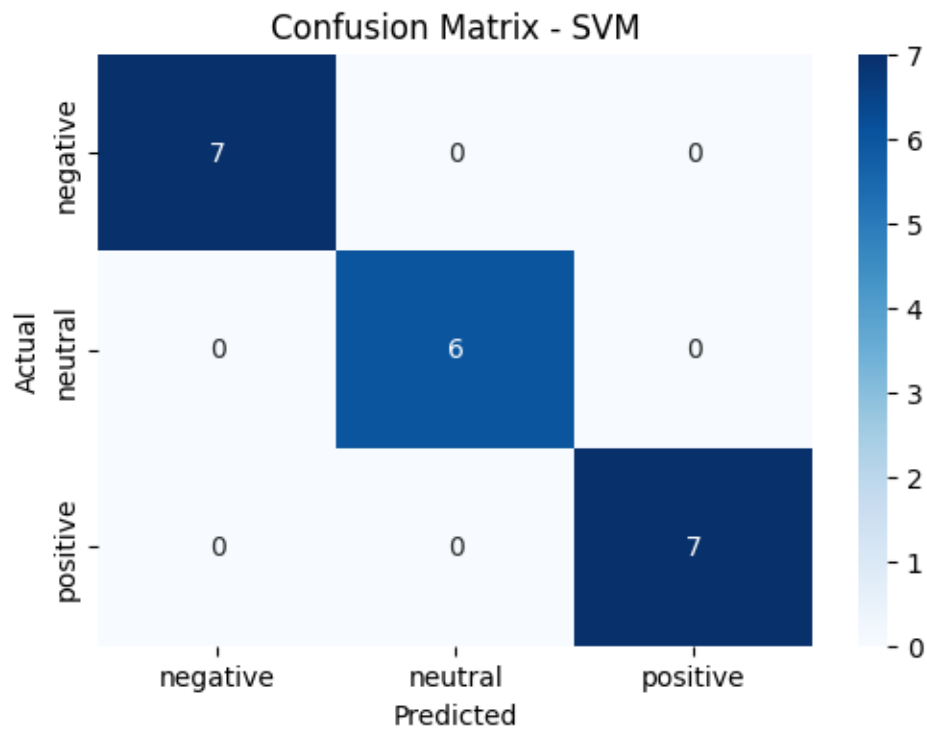
The performance of all three classifiers (Naive Bayes, SVM, and Logistic Regression) achieved perfect precision and recall scores of 1.00 on the test dataset.

This indicates that the models were able to classify all test samples correctly.

However, such perfect accuracy may be due to the small dataset size and the simplicity of the tweets, where sentiments were clearly distinguishable.

In real-world scenarios with larger and more complex datasets, the performance.





The following figures show the sentiment distribution, model comparison, and confusion matrix results obtained from the analysis.

CONCLUSION:

In this assignment, sentiment analysis was successfully performed on tweets related to World War 3 discussions. The dataset was manually labeled and analyzed using multiple machine learning models.

Among the models used, SVM showed the best performance in terms of precision and recall. This demonstrates the effectiveness of machine learning techniques in analyzing public sentiment from textual data.